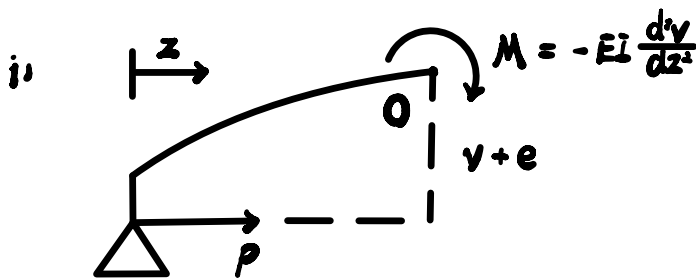
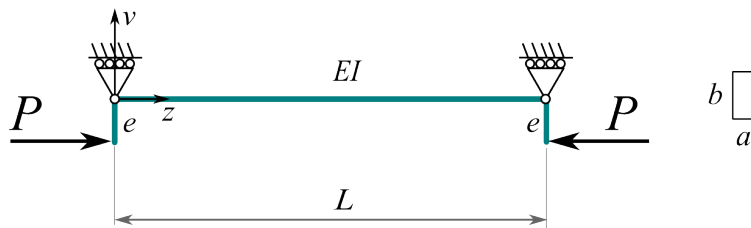


AERO50008 – Structures 2: Buckling of struts

*

- (1.) Assume the imperfect strut with eccentricity e , in the line of action of the load.
- Find its critical buckling load.
 - How does the eccentricity influence the buckling load?
 - What is the influence of the eccentricity upon the maximum deflection? Plot the ratio load P to critical Euler buckling ratio *vs.* maximum deflection.
 - Calculate the buckling load for a beam 10 m long, made of steel ($E = 210000$ MPa, $\nu = 0.3$), rectangular cross-section ($a = 100$ mm, $b = 400$ mm), with 100 mm eccentricity in the loading.



When reach the buckling load, at 0

$$M = -EI \frac{d^2v}{dz^2} = P(v + e) = Pv + Pe$$

$$\therefore EI \frac{d^2v}{dz^2} + Pv = -Pe \quad \therefore \frac{d^2v}{dz^2} + \mu^2 v = -\mu^2 e \quad \text{where } \mu^2 = \frac{P}{EI}$$

$$\therefore v = A \sin(\mu z) + B \cos(\mu z) + C$$

$$\therefore \frac{dv}{dz} = -\mu^2 [A \sin(\mu z) + B \cos(\mu z)]$$

$$\therefore \frac{dv}{dz} + \mu^2 v = \mu^2 C = -\mu^2 e \quad \therefore C = -e$$

$$\therefore v = A \sin(\mu z) + B \cos(\mu z) - e$$

$$\text{At } z = 0, v = 0$$

$$B - e = 0 \quad \therefore B = e \quad \therefore v = A \sin(\mu z) + e [\cos(\mu z) - 1]$$

$$\text{At } z = L, v = 0$$

$$\therefore A \sin(\mu L) + e [\cos(\mu L) - 1] = 0$$

$$\therefore A \neq 0 \quad \therefore \sin(\mu L) = \cos(\mu L) - 1 = 0 \quad \therefore \mu L = 0 \text{ or } n\pi \quad (n \in \mathbb{R})$$

$$\therefore \mu L = \pi \quad \therefore \mu^2 L^2 = \frac{P}{EI} L^2 = \pi^2 \quad \therefore P_{\text{crit}} = \frac{\pi^2 EI}{L^2}$$

ii) P_{crit} is not change

$$\text{iii) } A = \frac{-e [\cos(\mu L) - 1]}{\sin(\mu L)}$$

$$\text{At } z = \frac{L}{2},$$

$$v_c = \frac{-e[\cos(\frac{1}{2}\mu L) - 1]}{\sin(\frac{1}{2}\mu L)} \sin(\frac{1}{2}\mu L) + e \cos(\frac{1}{2}\mu L) - e$$

$$= \frac{2e \sin(\frac{1}{2}\mu L) \sin(\frac{1}{2}\mu L)}{2 \sin(\frac{1}{2}\mu L) \cos(\frac{1}{2}\mu L)} \sin(\frac{1}{2}\mu L) + e \cos(\frac{1}{2}\mu L) - e$$

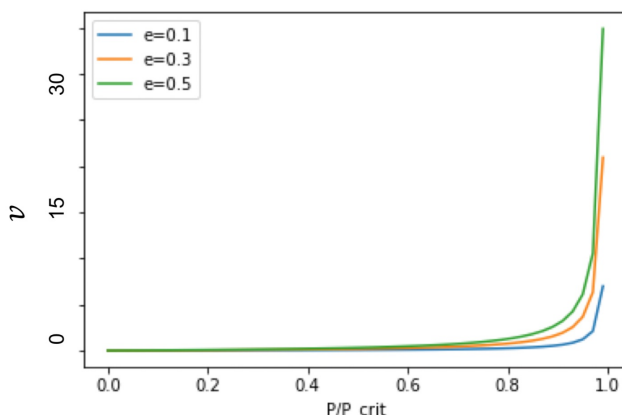
$$= e \frac{\sin^2(\frac{1}{2}\mu L)}{\cos(\frac{1}{2}\mu L)} + e \frac{\cos^2(\frac{1}{2}\mu L)}{\cos(\frac{1}{2}\mu L)} - e = e \left[\frac{1}{\cos(\frac{1}{2}\mu L)} - 1 \right]$$

$$\therefore \mu = \sqrt{\frac{P}{EI}} = \sqrt{\frac{P_{crit}}{EI} \frac{P}{P_{crit}}} = \sqrt{\frac{\pi^2 EI}{L^2} \cdot \frac{1}{EI}} \cdot \sqrt{\frac{P}{P_{crit}}} = \frac{\pi}{L} \sqrt{\frac{P}{P_{crit}}}$$

$$\therefore v_c = e \left[\frac{1}{\cos(\frac{\pi}{2} \sqrt{\frac{P}{P_{crit}}}} - 1 \right]$$

$\therefore v_c$ tend to be ∞ as $P_{crit} \rightarrow P$

$$\therefore \frac{P}{P_{crit}} = \left[\frac{2}{\pi} \cos^{-1} \left(\frac{e}{v_c + e} \right) \right]^2 = \left[\frac{2}{\pi} \cos^{-1} \left(\frac{e}{v_c} \right) \right]^2 \text{ for small } e$$

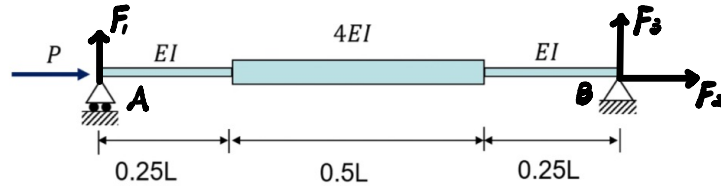


* ∞ at $P = P_{crit}$

$$iv) P_{crit} = \frac{\pi^2 EI}{L^2} = 1.1054 \times 10^8 \text{ N}$$

(2.) Consider the pin-ended strut shown below.

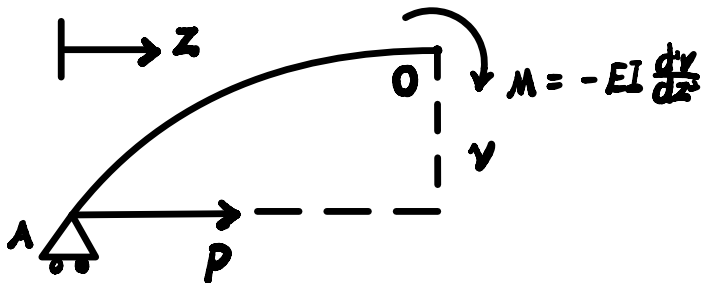
- Obtain the transcendental equation (i.e. displacement at any point) for the whole strut.
- Find the critical buckling load for the structure.



$$\sum F_v = F_1 + F_3 = 0$$

$$\sum F_h = F_2 + P = 0 \quad \therefore F_2 = -P$$

$$\text{At A, } M(A) = F_3 L = 0 \quad \therefore F_3 = 0 \quad \therefore F_1 = 0$$



$$\text{At B, } \sum M = -M + P v = EI \frac{dv}{dz} + P v = 0$$

$$\text{for } 0 \leq z \leq \frac{1}{4}L \text{ and } \frac{3}{4}L \leq z \leq L,$$

$$\therefore \frac{d^2 v}{dz^2} + \mu^2 v = 0 \quad \text{where } \mu^2 = \frac{P}{EI}$$

$$\therefore v_1 = A \sin(\mu z) + B \cos(\mu z)$$

for $\frac{1}{4}L \leq z \leq \frac{3}{4}L$.

$$\therefore \frac{d^2 v}{dz^2} + \frac{1}{4}\mu^2 v = 0 \quad \text{where } \mu^2 = \frac{P}{EI}$$

$$\therefore v_2 = C \sin\left(\frac{1}{2}\mu z\right) + D \cos\left(\frac{1}{2}\mu z\right)$$

$$\text{At } z=0, v_1=0$$

$$\therefore B=0 \quad \therefore v_1 = A \sin(\mu z)$$

$$\text{At } z = \frac{1}{2}L, \frac{dv_2}{dz} = 0$$

$$\therefore \frac{1}{2}\mu [C \cos\left(\frac{1}{4}\mu L\right) - D \sin\left(\frac{1}{4}\mu L\right)] = 0 \quad \therefore \mu \neq 0$$

$$\therefore C = D \tan\left(\frac{1}{4}\mu L\right)$$

$$\therefore v_2 = D \sin\left(\frac{1}{2}\mu z\right) \tan\left(\frac{1}{4}\mu L\right) + D \cos\left(\frac{1}{2}\mu z\right)$$

$$= D \left[\sin\left(\frac{1}{2}\mu z\right) \tan\left(\frac{1}{4}\mu L\right) + \cos\left(\frac{1}{2}\mu z\right) \right]$$

$$\text{At } z = \frac{1}{4}L, v_1 = v_2, \frac{dv_1}{dz} = \frac{dv_2}{dz}$$

$$\begin{aligned}
\therefore A \sin\left(\frac{1}{8}\mu L\right) &= D \left[\sin\left(\frac{1}{8}\mu L\right) \tan\left(\frac{1}{4}\mu L\right) + \cos\left(\frac{1}{8}\mu L\right) \right] \\
&= D \left[\sin\left(\frac{1}{8}\mu L\right) \frac{\sin\left(\frac{1}{4}\mu L\right)}{\cos\left(\frac{1}{4}\mu L\right)} + \cos\left(\frac{1}{8}\mu L\right) \right] \\
&= D \left[\frac{2 \sin^2\left(\frac{1}{8}\mu L\right) \cos\left(\frac{1}{8}\mu L\right)}{\cos\left(\frac{1}{4}\mu L\right)} + \cos\left(\frac{1}{8}\mu L\right) \right] \\
&= D \cos\left(\frac{1}{8}\mu L\right) \left[\frac{2 \sin^2\left(\frac{1}{8}\mu L\right) + \cos\left(\frac{1}{4}\mu L\right)}{\cos\left(\frac{1}{4}\mu L\right)} \right] \\
&= D \cos\left(\frac{1}{8}\mu L\right) \left[\frac{1 - \cos\left(\frac{1}{4}\mu L\right) + \cos\left(\frac{1}{4}\mu L\right)}{\cos\left(\frac{1}{4}\mu L\right)} \right] \\
&= D \cos\left(\frac{1}{8}\mu L\right) \sec\left(\frac{1}{4}\mu L\right)
\end{aligned}$$

$$\begin{aligned}
\mu A \cos\left(\frac{1}{4}\mu L\right) &= \frac{1}{2}\mu D \left[\cos\left(\frac{1}{8}\mu L\right) \tan\left(\frac{1}{4}\mu L\right) - \sin\left(\frac{1}{8}\mu L\right) \right] \\
&= \frac{1}{2}\mu D \left[\cos\left(\frac{1}{8}\mu L\right) \frac{2 \sin\left(\frac{1}{8}\mu L\right) \cos\left(\frac{1}{8}\mu L\right)}{\cos\left(\frac{1}{4}\mu L\right)} - \sin\left(\frac{1}{8}\mu L\right) \right] \\
&= \frac{1}{2}\mu D \sin\left(\frac{1}{8}\mu L\right) \left[\frac{2 \cos^2\left(\frac{1}{8}\mu L\right) - \cos\left(\frac{1}{4}\mu L\right)}{\cos\left(\frac{1}{4}\mu L\right)} \right] \\
&= \frac{1}{2}\mu D \sin\left(\frac{1}{8}\mu L\right) \left[\frac{1 + \cos\left(\frac{1}{4}\mu L\right) - \cos\left(\frac{1}{4}\mu L\right)}{\cos\left(\frac{1}{4}\mu L\right)} \right] \\
&= \frac{1}{2}\mu D \sin\left(\frac{1}{8}\mu L\right) \sec\left(\frac{1}{4}\mu L\right)
\end{aligned}$$

$$\begin{aligned}\therefore A &= D \cos\left(\frac{1}{8}\mu L\right) \sec\left(\frac{1}{4}\mu L\right) \operatorname{cosec}\left(\frac{1}{4}\mu L\right) \\ &= \frac{1}{2} D \sin\left(\frac{1}{8}\mu L\right) \sec^2\left(\frac{1}{4}\mu L\right)\end{aligned}$$

$$\therefore \cos\left(\frac{1}{8}\mu L\right) \operatorname{cosec}\left(\frac{1}{4}\mu L\right) = \frac{1}{2} \sin\left(\frac{1}{8}\mu L\right) \sec\left(\frac{1}{4}\mu L\right)$$

$$\therefore \frac{1}{2} = \frac{1}{\tan\left(\frac{1}{8}\mu L\right) \cdot \tan\left(\frac{1}{4}\mu L\right)}$$

$$\therefore \frac{1}{2} \tan\left(\frac{1}{4}\mu L\right) = \frac{1}{\tan\left(\frac{1}{8}\mu L\right)}$$

$$\therefore \tan(2x) = \frac{2 \tan(x)}{1 - \tan^2(x)}$$

$$\therefore \tan\left(\frac{1}{4}\mu L\right) = \frac{2 \tan\left(\frac{1}{8}\mu L\right)}{1 - \tan^2\left(\frac{1}{8}\mu L\right)}$$

$$\therefore \frac{\tan\left(\frac{1}{8}\mu L\right)}{1 - \tan^2\left(\frac{1}{8}\mu L\right)} = \frac{1}{\tan\left(\frac{1}{8}\mu L\right)}$$

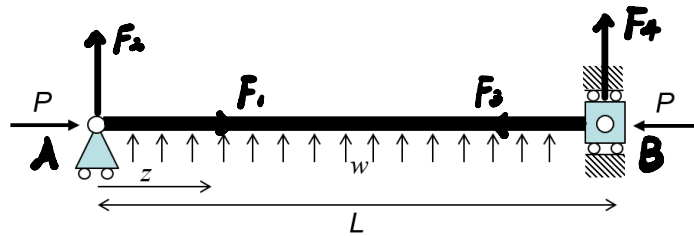
$$\therefore \tan^2\left(\frac{1}{8}\mu L\right) = \frac{1}{2} \quad \therefore \tan\left(\frac{1}{8}\mu L\right) = \frac{1}{\sqrt{2}} \quad \therefore \frac{1}{8}\mu L = 0.6155$$

$$\therefore \mu L = 4.9238 \quad \therefore \mu^2 L^2 = \frac{P}{EI} L^2 = 24.2442$$

$$\therefore P_{crit} = \frac{24.2442 EI}{L^2}$$

(3.) Consider the pin-ended strut shown in the figure, subjected to an axial load P and a uniformly distributed lateral force of intensity w . Derive the expressions for the mid-span deflection and the critical buckling load, and discuss how this is influenced by the lateral load.

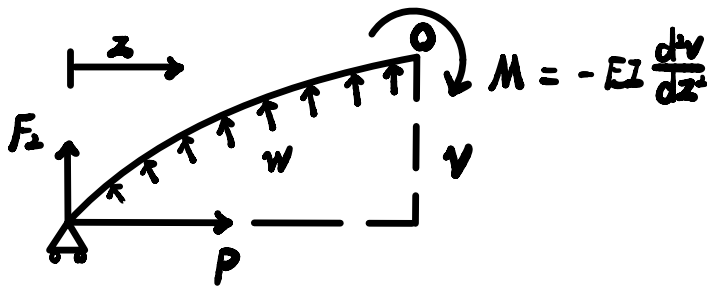
- Formulate the differential equation to assess stability of the system.
- Find the critical buckling load for the structure.



$$\sum F_v = F_2 + F_4 = -wL$$

$$\sum F_h = F_1 + F_3 = 0 \quad \therefore F_1 = -F_3 = 0$$

$$M(A), F_4 L + \frac{1}{2} w L^2 = 0 \quad \therefore F_4 = -\frac{1}{2} w L \quad \therefore F_2 = -\frac{1}{2} w L$$



$$\text{At } 0, \sum M = EI \frac{d^2v}{dz^2} - \frac{1}{2} w z^2 + P v + \frac{1}{2} w L z = 0$$

$$\therefore EI \frac{d^2 v}{dz^2} + Pv = \frac{1}{2} w z^2 - \frac{1}{2} w L z$$

$$\therefore \frac{d^2 v}{dz^2} + \mu^2 v = \frac{w z^2}{2EI} - \frac{w L z}{2EI} \quad \text{where } \mu^2 = \frac{P}{EI}$$

$$\therefore v = A \sin(\mu z) + B \cos(\mu z) + C z^2 + D z + E$$

$$\therefore \frac{d^2 v}{dz^2} = -\mu^2 [A \sin(\mu z) + B \cos(\mu z)] + 2C$$

$$\therefore \mu^2 C z^2 + \mu^2 D z + \mu^2 E + 2C = \frac{w z^2}{2EI} - \frac{w L z}{2EI}$$

$$\therefore C = \frac{w}{2EI\mu^2} = \frac{w}{2P}$$

$$D = \frac{-wL}{2EI\mu^2} = -\frac{wL}{2P}$$

$$E = -\frac{2C}{\mu^2} = -\frac{w}{P\mu^2}$$

$$\therefore v = A \sin(\mu z) + B \cos(\mu z) + \frac{w}{2P} z^2 - \frac{wL}{2P} z - \frac{w}{P\mu^2}$$

$$\text{At } z=0, v=0$$

$$\therefore B - \frac{w}{P\mu^2} = 0 \quad \therefore B = \frac{w}{P\mu^2}$$

$$\text{At } z = L, v = 0$$

$$\therefore A \sin(\mu L) + \frac{W}{P\mu^2} \cos(\mu L) - \frac{W}{P\mu^2} = 0$$

$$\therefore A \sin(\mu L) + \frac{W}{P\mu^2} [\cos(\mu L) - 1] = 0$$

$$\therefore A \neq 0 \quad \therefore A = -\frac{W}{P\mu^2} \frac{\cos(\mu L) - 1}{\sin(\mu L)}$$

$$\therefore \sin(\mu L) = \cos(\mu L) - 1 = 0$$

$$\therefore \mu L = 0 \text{ or } n\pi \quad (n \in \mathbb{R}) \quad \therefore \mu L = \pi$$

$$\therefore \mu^2 L^2 = \pi^2 = \frac{P}{EI} L^2$$

$$\therefore P_{crit} = \frac{\pi^2 EI}{L^2}$$

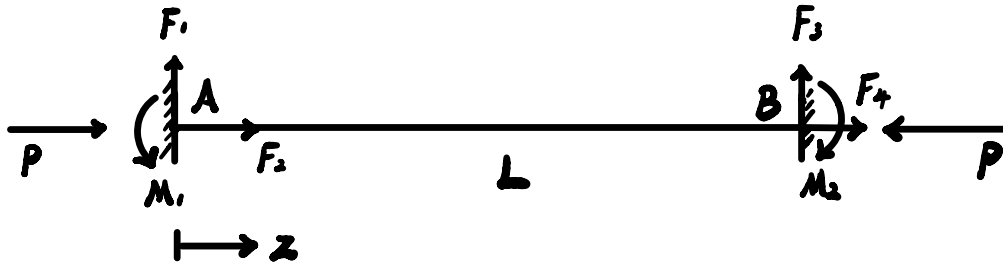
$$\text{At } z = \frac{L}{2}, v_c = v_{max}$$

$$\begin{aligned} \therefore v_c &= -\frac{W}{P\mu^2} \frac{\cos(\mu L) - 1}{\sin(\mu L)} \sin\left(\frac{1}{2}\mu L\right) + \frac{W}{P\mu^2} \cos\left(\frac{1}{2}\mu L\right) + \frac{W}{8P} L^2 - \frac{WL}{4P} L - \frac{W}{P\mu^2} \\ &= \frac{W}{P\mu^2} \left[\frac{2\sin^2\left(\frac{1}{2}\mu L\right)}{2\sin\left(\frac{1}{2}\mu L\right)\cos\left(\frac{1}{2}\mu L\right)} \sin\left(\frac{1}{2}\mu L\right) + \cos\left(\frac{1}{2}\mu L\right) - 1 \right] - \frac{WL}{8P} L^2 \end{aligned}$$

$$\therefore v_c = \frac{W}{\rho \mu^2} \left[\frac{\sinh(\frac{1}{2}\mu L) + \cosh(\frac{1}{2}\mu L)}{\cosh(\frac{1}{2}\mu L)} - 1 \right] - \frac{W}{8\rho} L^2$$

$$= \frac{W}{\rho \mu^2} \left[\sec(\frac{1}{2}\mu L) - 1 \right] - \frac{W}{8\rho} L^2$$

(4.) Calculate the buckling load of a fixed-fixed strut.



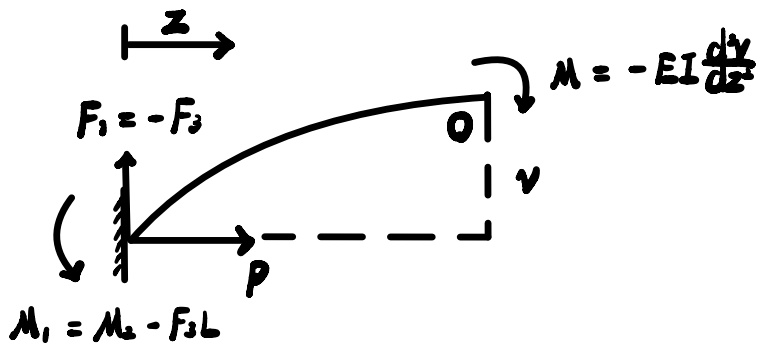
$$\sum F_v = F_1 + F_3 = 0 \quad \therefore F_1 = -F_3$$

$$\sum F_h = F_2 + F_4 = 0 \quad \therefore F_2 = F_4 = 0$$

$$\text{At } A, \sum M = M_1 + F_3 L - M_2 = 0$$

$$\text{At } B, \sum M = M_2 + F_1 L - M_1 = 0$$

$$\therefore M_1 = M_2 - F_3 L$$



$$\text{At } O, \sum M = EI \frac{d^2v}{dz^2} + P v + F_3 z + M_2 - F_3 L$$

$$\therefore \frac{d^2v}{dz^2} + \mu^2 v = \frac{F_3}{EI} (L - z) - \frac{M_2}{EI} \quad \text{where } \mu^2 = \frac{P}{EI}$$

$$\therefore v = A \sin(\mu z) + B \cos(\mu z) + Cz + D$$

$$\therefore \frac{d^4 v}{dz^4} = -\mu^4 [A \sin(\mu z) + B \cos(\mu z)]$$

$$\therefore \frac{d^4 v}{dz^4} + \mu^4 v = \mu^4 Cz + \mu^4 D = \frac{F_3}{EI}(L-z) - \frac{M_2}{EI} = -\frac{F_3}{EI}z + \frac{F_3}{EI}L - \frac{M_2}{EI}$$

$$\therefore C = -\frac{F_3}{EI\mu^2} = -\frac{F_3}{P}$$

$$D = \frac{F_3}{EI\mu^2}L - \frac{M_2}{EI\mu^2} = \frac{F_3}{P}L - \frac{M_2}{P}$$

$$\therefore v = A \sin(\mu z) + B \cos(\mu z) - \frac{F_3}{P}z + \frac{F_3}{P}L - \frac{M_2}{P}$$

$$\text{At } z=0, v=0 \text{ and } \frac{dv}{dz}=0$$

$$\therefore B + \frac{F_3}{P}L - \frac{M_2}{P} = 0 \text{ and } \mu A - \frac{F_3}{P} = 0$$

$$\therefore B = -\frac{F_3}{P}L + \frac{M_2}{P} \text{ and } A = \frac{F_3}{P\mu}$$

$$\therefore v = \frac{F_3}{P\mu} \sin(\mu z) + \left(\frac{M_2}{P} - \frac{F_3}{P}L \right) \cos(\mu z) - \frac{F_3}{P}z + \frac{F_3}{P}L - \frac{M_2}{P}$$

$$\text{At } z=L, v=0 \text{ and } \frac{dv}{dz}=0$$

$$\therefore \frac{F_3}{P\mu} \sin(\mu L) + \left(\frac{M_2}{P} - \frac{F_3}{P} L \right) [\cos(\mu L) - 1] - \frac{F_2}{P} L = 0$$

$$\text{and } \frac{F_3}{P} \cos(\mu L) - \mu \left(\frac{M_2}{P} - \frac{F_3}{P} L \right) \sin(\mu L) - \frac{F_2}{P} = 0$$

$$\therefore F_1 = -F_3 = 0$$

$$\therefore \frac{M_2}{P} [\cos(\mu L) - 1] = 0 \quad \text{and} \quad \mu \frac{M_2}{P} \sin(\mu L) = 0$$

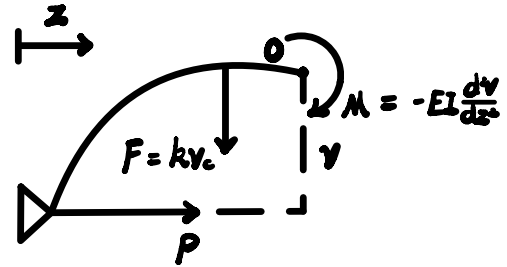
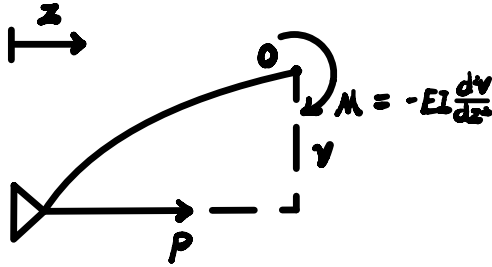
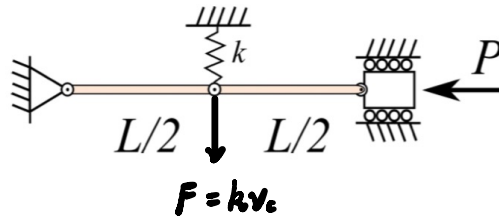
$$\therefore \sin(\mu L) = 0 \quad \text{while} \quad \cos(\mu L) = 1$$

$$\therefore \mu L = 0 \text{ or } 2n\pi, \quad n \in \mathbb{R} \qquad \therefore \mu L = 2\pi$$

$$\therefore \mu^2 L^2 = 4\pi^2 = \frac{P}{EI} L^2$$

$$\therefore P = \frac{4\pi^2 EI}{L^2}$$

(5.) The structure shown below consists of two rigid members of length $L/2$ connected via pins. Use k as the spring's stiffness constant. What is the critical buckling load?



for $0 \leq z \leq \frac{1}{2}L$, $\mu^2 = \frac{P}{EI}$

$$\sum M(0) = EI \frac{d^2v}{dz^2} + Pv = 0 \quad \therefore \frac{d^2v}{dz^2} + \mu^2 v = 0$$

$$\therefore v_1 = A \sin(\mu z) + B \cos(\mu z)$$

for $\frac{1}{2}L \leq z \leq L$, $\mu^2 = \frac{P}{EI}$

$$\sum M(0) = EI \frac{d^2v}{dz^2} + Pv + kv_c(z - \frac{1}{2}L) = 0 \quad \therefore \frac{d^2v}{dz^2} + \mu^2 v = \frac{kv_c}{EI}(\frac{1}{2}L - z)$$

$$\therefore v_2 = C \sin(\mu z) + D \cos(\mu z) + Ez + F$$

$$\therefore \frac{d^2v_2}{dz^2} = -\mu^2 [C \sin(\mu z) + D \cos(\mu z)]$$

$$\therefore Ez + F = \frac{kV_0}{EI} \left(\frac{1}{2}L - z \right)$$

$$\therefore v_2 = C \sin(\mu z) + D \cos(\mu z) - \frac{kV_0}{EI} z + \frac{kV_0}{2EI} L$$

$$\text{At } z = 0, v = 0$$

$$\therefore v_1 = B = 0 \quad \therefore v_1 = A \sin(\mu z)$$

$$\text{At } z = \frac{1}{2}L, v_1 = v_2, \quad \frac{dv_1}{dz} = \frac{dv_2}{dz}$$

$$\therefore A \sin\left(\frac{1}{2}\mu L\right) = C \sin\left(\frac{1}{2}\mu L\right) + D \cos\left(\frac{1}{2}\mu L\right)$$

$$\mu A \cos\left(\frac{1}{2}\mu L\right) = \mu C \cos\left(\frac{1}{2}\mu L\right) - \mu D \sin\left(\frac{1}{2}\mu L\right) - \frac{kV_0}{EI}$$

$$\therefore C + D \cot\left(\frac{1}{2}\mu L\right) = C - D \tan\left(\frac{1}{2}\mu L\right) - \frac{kV_0}{EI\mu} \sec\left(\frac{1}{2}\mu L\right)$$

$$\therefore D \left[\cot\left(\frac{1}{2}\mu L\right) + \tan\left(\frac{1}{2}\mu L\right) \right] + \frac{kV_0}{EI\mu} \sec\left(\frac{1}{2}\mu L\right) = 0$$

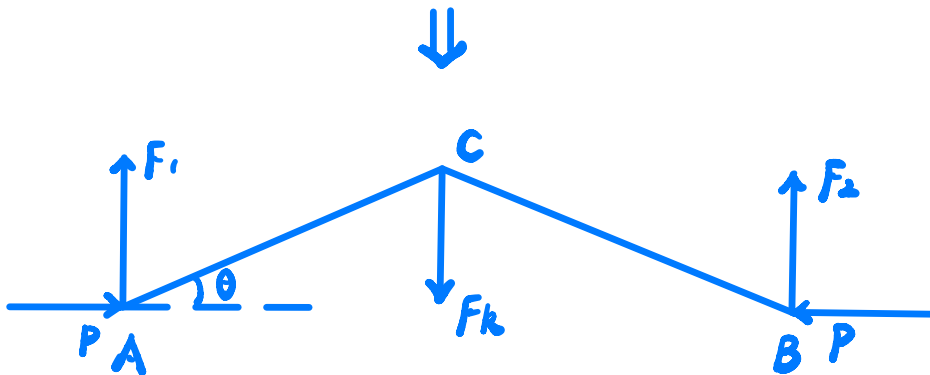
$$\text{At } z = L, v_2 = 0$$

$$\therefore C \sin(\mu L) + D \cos(\mu L) - \frac{kV_0}{2EI} L = 0$$

$$\Sigma F_v = F_1 + F_2 = F_k$$

$$\Sigma M_A = F_2 L - F_k \left(\frac{1}{2}L\right) = 0 \quad \therefore F_2 = \frac{1}{2}F_k = F_1$$

$$F_k = k \cdot L = k \cdot y_c = k \cdot \frac{1}{2}L \sin \theta = \frac{1}{2}kL\theta$$



$$\text{At } A, \Sigma F_v = F_1 + T_{AC} \sin \theta = 0$$

$$\Sigma F_h = P + T_{AC} \cos \theta = 0$$

$$\therefore F_1 = P \tan \theta = P\theta = \frac{1}{2}F_k = \frac{1}{4}kL\theta$$

$$\therefore P = \frac{1}{4}kL$$

(6.) A compressive load P is applied aligned with the axis of a column of length 1.4 m with pinned ends. The column has a constant rectangular cross section of dimensions $d \times 3d$. Calculate the smallest possible value for d if $P = 108$ kN. The material properties are: $E = 200$ GPa, $\sigma_y = 247$ MPa. Consider a factor of safety (FS) of 1.92. $FS = \frac{\text{failure stress}}{\text{actual stress}}$.

Assume $R_{\text{trans}} \leq \frac{L}{r}$, fully buckling

$$\sigma_{\text{actual}} = \frac{P_{\text{crit}}}{A} = \frac{P_{\text{crit}}}{3d^2}$$

$$\therefore r = \sqrt{\frac{I}{A}} = \sqrt{\frac{\frac{3d^4}{12}}{3d^2}} = \sqrt{\frac{d^2}{12}} = \frac{\sqrt{3}}{6}d$$

$$\therefore \sigma_{\text{crit}} = \frac{P_{\text{crit}}}{A} = \frac{\pi^2 EI}{L^2} \frac{1}{A} = \pi^2 E \left(\frac{r}{L}\right)^2 = \frac{\pi^2 E}{\left(\frac{L}{r}\right)^2} = \frac{\pi^2 E d^2}{12L^2}$$

$$\therefore \frac{\sigma_{\text{crit}}}{\sigma_{\text{actual}}} = 1.92 \quad \therefore \frac{\pi^2 E d^2}{12L^2} = \frac{1.92 P_{\text{crit}}}{3d^2} \quad \therefore d = 0.03013 \text{ m}$$

$$\therefore \frac{L}{r} = 161$$

$$\therefore R_{\text{trans}} = \left(\frac{2\pi^2 E}{6Y} \right)^{\frac{1}{2}} = 126.42$$

$$\therefore \frac{L}{r} > R_{\text{trans}}, \text{ assumption is correct}$$

Answers

1. Shown in video on Panopto (link on BB).

$$P_{crit} = \frac{\pi^2 EI}{L^2}$$

$$\frac{P}{P_{crit}} = \left(\frac{2}{\pi} \cos^{-1}\left(\frac{e}{v_c}\right)\right)^2$$

- 2.

$$v_{AB} = B \sin(\mu z)$$

$$v_{BC} = C \left(\cos\left(\frac{\mu z}{2}\right) + \tan\left(\frac{\mu z}{4}\right) \sin\left(\frac{\mu z}{2}\right) \right)$$

$$P_{crit} = \frac{24.2EI}{L^2}$$

- 3.

$$\frac{d^2 v}{dz^2} + \mu^2 v = -\frac{wLz}{2EI} \left(z^2 - zL - \frac{2}{\mu^2} \right)$$

$$P_{crit} = \frac{\pi^2 EI}{L^2}$$

- 4.

$$P_{crit} = \frac{\pi^2 EI}{(0.5L)^2}$$

- 5.

$$P_{crit} = \frac{1}{4} kL$$

- 6.

$$d = 0.0301 \text{m}$$